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Java SE 27 & JDK 27

Class Enum<E extends Enum<E>>

java.lang.Object
    java.lang.Enum<E>

Type Parameters:

E - The type of the enum subclass

All Implemented Interfaces:

Serializable, Comparable<E>, Constable

```
public abstract class Enum<E extends Enum<E>>
extends Object
implements Constable, Comparable<E>, Serializable
```

This is the common base class of all Java language enumeration classes. More information about enums, including descriptions of the implicitly declared methods synthesized by the compiler, can be found in section §8.9 of *The Java Language Specification*. Enumeration classes are all serializable and receive special handling by the serialization mechanism. The serialized representation used for enum constants cannot be customized. Declarations of methods and fields that would otherwise interact with serialization are ignored, including `serialVersionUID`; see the *Java Object Serialization Specification* for details.

Note that when using an enumeration type as the type of a set or as the type of the keys in a map, specialized and efficient set and map implementations are available.

See *Java Language Specification*:

8.9 Enum Classes
8.9.3 Enum Members

Since:

1.5

External Specifications

Java Object Serialization Specification

See Also:

Class.getEnumConstants(), EnumSet, EnumMap

Nested Class Summary

Nested Classes

Modifier and Type	Class	Description
static final class	**Enum.EnumDesc**<E extends Enum<E>>	A nominal descriptor for an enum constant.

Constructor Summary

Constructors

Modifier	Constructor	Description
protected	`Enum(String name, int ordinal)`	Sole constructor.

Method Summary

All Methods | **Static Methods**

Modifier and Type	Method	Description
protected final Object	`clone()`	Throws CloneNotSupportedException.
final int	`compareTo(E o)`	Compares this enum with the specified object for order.
final Optional<Enum.EnumDesc<E>>	`describeConstable()`	Returns an enum descriptor `EnumDesc` for this instance, if one can be constructed, or an empty `Optional` if one cannot be.
final boolean	`equals(Object other)`	Returns true if the specified object is equal to this enum constant.
protected final void	`finalize()`	**Deprecated, for removal: This API element is subject to removal in a future version.** Finalization has been deprecated for removal.
final Class<E>	`getDeclaringClass()`	Returns the Class object corresponding to this enum constant's enum type.
final int	`hashCode()`	

		Returns a hash code for this enum constant.
final String	**name**()	Returns the name of this enum constant, exactly as declared in its enum declaration.
final int	**ordinal**()	Returns the ordinal of this enumeration constant (its position in its enum declaration, where the initial constant is assigned an ordinal of zero).
String	**toString**()	Returns the name of this enum constant, as contained in the declaration.
static <T extends Enum<T>> T	**valueOf**(Class<T> enumClass, String name)	Returns the enum constant of the specified enum class with the specified name.

Methods declared in class Object

getClass, notify, notifyAll, wait, wait, wait

Constructor Details

Enum

```
protected Enum(String name,
               int ordinal)
```

Sole constructor. Programmers cannot invoke this constructor. It is for use by code emitted by the compiler in response to enum class declarations.

Parameters:

`name` - The name of this enum constant, which is the identifier used to declare it.

`ordinal` - The ordinal of this enumeration constant (its position in the enum declaration, where the initial constant is assigned an ordinal of zero).

Method Details

name

```
public final String name()
```

Returns the name of this enum constant, exactly as declared in its enum declaration. **Most programmers should use the `toString()` method in preference to this one, as the toString method may return a more user-friendly name.** This method is designed primarily for use in specialized situations where correctness depends on getting the exact name, which will not vary from release to release.

Returns:

the name of this enum constant

ordinal

```
public final int ordinal()
```

Returns the ordinal of this enumeration constant (its position in its enum declaration, where the initial constant is assigned an ordinal of zero). Most programmers will have no use for this method. It is designed for use by sophisticated enum-based data structures, such as `EnumSet` and `EnumMap`.

Returns:

the ordinal of this enumeration constant

toString

```
public String toString()
```

Returns the name of this enum constant, as contained in the declaration. This method may be overridden, though it typically isn't necessary or desirable. An enum class should override this method when a more "programmer-friendly" string form exists.

Overrides:

`toString` in class `Object`

Returns:

the name of this enum constant

equals

```
public final boolean equals(Object other)
```

Returns true if the specified object is equal to this enum constant.

Overrides:

`equals` in class `Object`

Parameters:

`other` - the object to be compared for equality with this object.

Returns:

true if the specified object is equal to this enum constant.

See Also:

`Object.hashCode()`, `HashMap`

hashCode

```
public final int hashCode()
```

Returns a hash code for this enum constant.

Overrides:

`hashCode` in class `Object`

Returns:

a hash code for this enum constant.

See Also:

`Object.equals(java.lang.Object)`,
`System.identityHashCode(Object)`

clone

```
protected final Object clone()
                      throws CloneNotSupportedException
```

Throws CloneNotSupportedException. This guarantees that enums are never cloned, which is necessary to preserve their "singleton" status.

Overrides:

`clone` in class `Object`

Returns:

(never returns)

Throws:

`CloneNotSupportedException` - if the object's class does not support the `Cloneable` interface. Subclasses that override the `clone` method can also throw this exception to indicate that an instance cannot be cloned.

See Also:

`Cloneable`

compareTo

```
public final int compareTo(E o)
```

Compares this enum with the specified object for order. Returns a negative integer, zero, or a positive integer as this object is less than, equal to, or greater than the specified object. Enum constants are only comparable to other enum constants of the same enum type. The natural order implemented by this method is the order in which the constants are declared.

Specified by:

`compareTo` in interface `Comparable<E extends Enum<E>>`

Parameters:

o - the object to be compared.

Returns:

a negative integer, zero, or a positive integer as this object is less than, equal to, or greater than the specified object.

getDeclaringClass

```
public final Class<E> getDeclaringClass()
```

Returns the Class object corresponding to this enum constant's enum type. Two enum constants e1 and e2 are of the same enum type if and only if e1.getDeclaringClass() == e2.getDeclaringClass(). (The value returned by this method may differ from the one returned by the `Object.getClass()` method for enum constants with constant-specific class bodies.)

Returns:

the Class object corresponding to this enum constant's enum type

describeConstable

```
public final Optional<Enum.EnumDesc<E>> describeConstable()
```

Returns an enum descriptor `EnumDesc` for this instance, if one can be constructed, or an empty `Optional` if one cannot be.

Specified by:

`describeConstable` in interface `Constable`

Returns:

An `Optional` containing the resulting nominal descriptor, or an empty `Optional` if one cannot be constructed.

Since:

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valueOf

```
public static <T extends Enum<T>> T valueOf(Class<T> enumClass,
                                            String name)
```

Returns the enum constant of the specified enum class with the specified name. The name must match exactly an identifier used to declare an enum constant in this class. (Extraneous whitespace characters are not permitted.)

Note that for a particular enum class `T`, the implicitly declared `public static T valueOf(String)` method on that enum may be used instead of this method to map from a name to the corresponding enum constant. All the constants of an enum class can be obtained by calling the implicit `public static T[] values()` method of that class.

Type Parameters:

`T` - The enum class whose constant is to be returned

Parameters:

`enumClass` - the `Class` object of the enum class from which to return a constant

`name` - the name of the constant to return

Returns:

the enum constant of the specified enum class with the specified name

Throws:

`IllegalArgumentException` - if the specified enum class has no constant with the specified name, or the specified class object does not represent an enum class

`NullPointerException` - if `enumClass` or `name` is null

Since:

1.5

finalize

```
@Deprecated(since="18",
            forRemoval=true)
protected final void finalize()
```

Deprecated, for removal: This API element is subject to removal in a future version.
Finalization has been deprecated for removal. See `Object.finalize()` for background information and details about migration options.

enum classes cannot have finalize methods.

Overrides:

`finalize` in class `Object`

See Also:

`WeakReference`, `PhantomReference`

Report a bug or suggest an enhancement
For further API reference and developer documentation see the Java SE Documentation, which contains more detailed, developer-targeted descriptions with conceptual overviews, definitions of terms, workarounds, and working code examples. Other versions.
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